

Leeds Bus Reliability and Delay Analytics API

- ▶ Project goal: analyse Leeds bus reliability and delay patterns.
- ▶ Tech stack: FastAPI + SQLite.
- ▶ Data source: UK Bus Open Data Service (BODS).
- ▶ Core capability: read endpoints + incident write endpoints + reliability analytics.

Keywords

FastAPI **SQLite** **BODS**

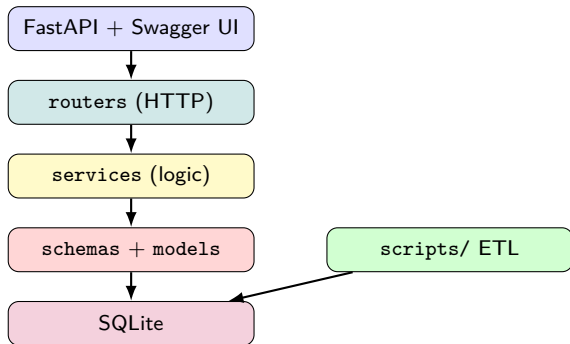
Public transport API **Bus reliability** **Delay analytics** **Leeds**

Why is the scope intentionally narrowed?

To keep implementation feasible, demonstrable, and testable within coursework constraints while preserving real-data value.

- | | |
|---------------------|---|
| Leeds only | A focused geography keeps ETL volume and validation complexity manageable. |
| BODS only | Single data source avoids cross-provider ID mismatch and data model conflicts. |
| 10 endpoints | Enough to demonstrate transport query + CRUD + analytics without over-engineering. |
| X-API-Key | Lightweight authentication suitable for coursework; avoids full JWT user system overhead. |

System Architecture

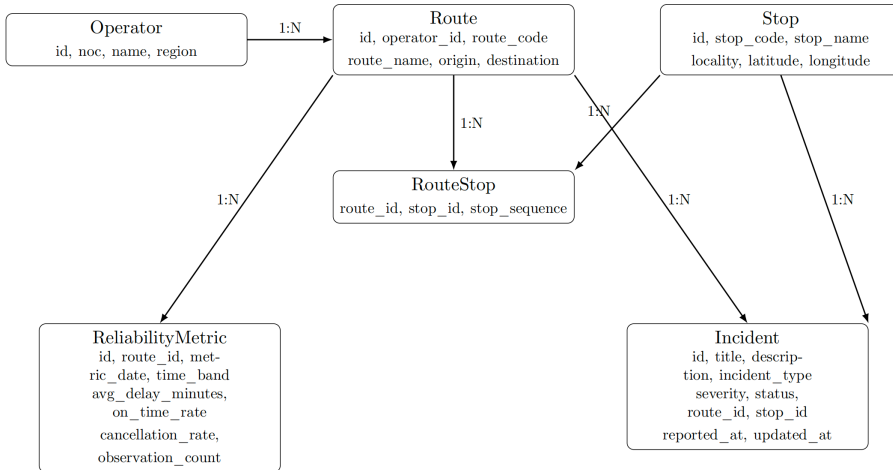


- ▶ FastAPI provides endpoints and Swagger UI.
- ▶ SQLite stores route, stop, incident, and metric data.
- ▶ ETL scripts ingest and standardise BODS data.

Keywords

Layered architecture Routers Services Schemas Models SQLite Swagger UI

Database Structure

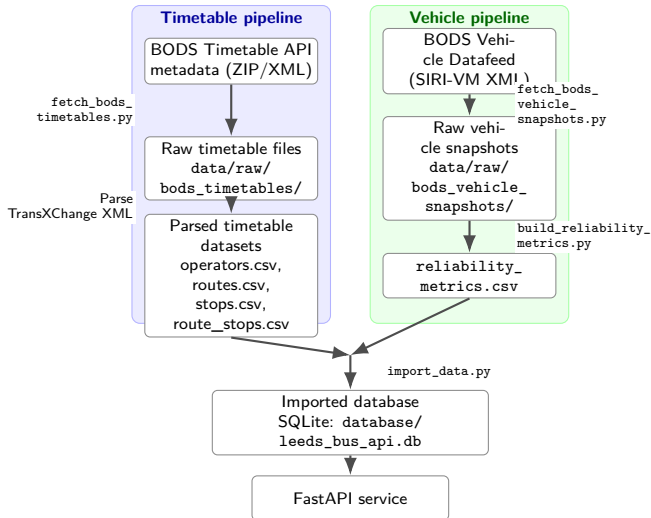


Data Pipeline and Real Data Processing

- ▶ `fetch_bods_timetables.py`
- ▶ `fetch_bods_vehicle_snapshots.py`
- ▶ `build_reliability_metrics.py`
- ▶ `import_data.py`

Processing flow

Raw XML/ZIP (TransXChange + SIRI-VM)
→ processed CSV → SQLite



Access control

GET endpoints are public; POST/PUT/DELETE endpoints require X-API-Key.

Implemented Endpoints

GET /operators	GET /routes
GET /routes/{route_id}	GET /stops
POST /incidents	GET /incidents
GET	PUT
/incidents/{incident_id}	/incidents/{incident_id}
DELETE	
/incidents/{incident_id}	
GET /analytics/routes/{route_id}/reliability	

HTTP Status Mapping

200 OK	successful reads and updates
201 Created	incident creation
204 No Content	successful deletion
401 Unauthorized	missing/invalid API keys
404 Not Found	missing routes or incidents
422 Unprocessable Entity	request validation errors

Results, Limitations and Next Steps

Completed

- ▶ Real BODS data import
- ▶ SQLite persistence
- ▶ Route, stop, incident, reliability APIs
- ▶ Swagger + PDF documentation

Limitations

- ▶ Demo-sized dataset
- ▶ Basic analytics indicators
- ▶ No complex user system/JWT

Future Work

- ▶ Larger data volume
- ▶ Finer time-band analysis
- ▶ Frontend dashboard visualisation

Keywords

Completed Real data import Analytics endpoint Limitations
Demo dataset Future work Scalability